

CHAROTARUNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Pharm. Examination

University Backlog Theory Examination May 2018

PH210 Organic Chemistry-II [Old Syllabus]

Date: 14/05/2018, Monday

Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

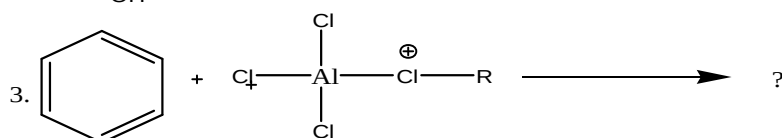
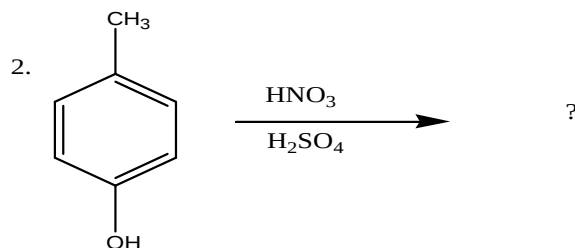
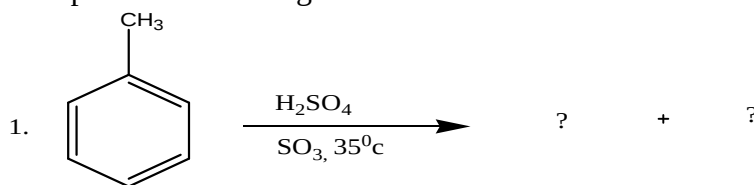
Q. 1 Attempt any TWO of the following:

- | | | |
|----------|---|-----------|
| A | 1. Explain Sulphonation reaction of Arenes. | 05 |
| | 2. Give any three preparation of Ketone. | 05 |
| B | 1. Reaction and mechanism of Diel's alder reaction. | 05 |
| | 2. Explain the different reaction of Naphthalene. | 05 |
| C | 1. Give the Preparation of Carboxylic Acid. | 05 |
| | 2. Give any three reactions of aromatic primary amines. | 05 |

Q. 2 Attempt any TWO of the following:

- | | | |
|----------|---|-----------|
| A | 1. Give any three preparations of Ester. | 05 |
| | 2. Give any three reactions of Phenol. | 05 |
| B | 1. Write a short note on Friedel craft alkylation with mechanism and suitable examples. | 05 |
| | 2. Explain following terms with suitable examples; (Any Five) | 05 |
| | i. Eclipsed and Staggered | |
| | ii. Optical Isomers | |
| | iii. Erythro and Threo | |
| | iv. E and Z Configuration | |
| | v. Relative Configuration | |
| | vi. Chirality | |

C 1. Complete the following reactions: 06



2. Give structures and Chemical Name of (i) Napthalene (ii) Benzoic Acid 04

SECTION- II

Q. 3 **Attempt any TWO of the following:**

A 1. What different conditions are to be fulfilled by a compound to show Aromaticity. 05

2. Explain the reaction of acetyl chloride with toluene in presence of aluminum chloride, what is the name of this Name reaction? 05

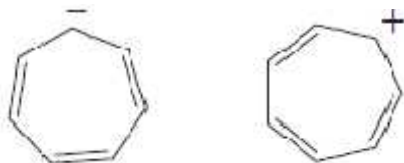
B 1. What are ring activator and ring deactivator towards Electrophilic Aromatic Substitution? Why halogens are o,p-directors even though they are electron withdrawing groups? 05

2. Write a note on Sigma tropic rearrangements. 05

C What is Green Chemistry? Explain the twelve principle of Green Chemistry. 10

Q. 4 **Attempt any TWO of the following:**

A 1. Which of the following compound you expect to be aromatic? Justify your answer. 05



2. Write a note on Electro cyclic reaction. 05

B What is Electrophilic Aromatic Substitution ? Explain the halogenation of aromatic compound with suitable example. 10

C 1. Write a note on microwave synthesis. Give the merits and demerits of microwave synthesis. 10